

항공기계공학과 2022010491 오서준

전산유체해석실습 11주차 보고서



해석과제

rot_Caradonna 해석 및 실험값과 비교

해석조건

1. Caradonna & Tung rotor의 단면 압력분포를 실험데이터와 비교하여 정리하시오.
 - 실험값과 단면 위치는 아래 표를 참고하시오
 - 해석조건은 마하수 0.877, 회전RPM 2,500, 피치각 8도를 적용하시오

TABLE 25.- LOCAL PRESSURE COEFFICIENT

$\theta_c = 8^\circ$ $\Omega = 2500$ rpm $M_{tip} = 0.877$

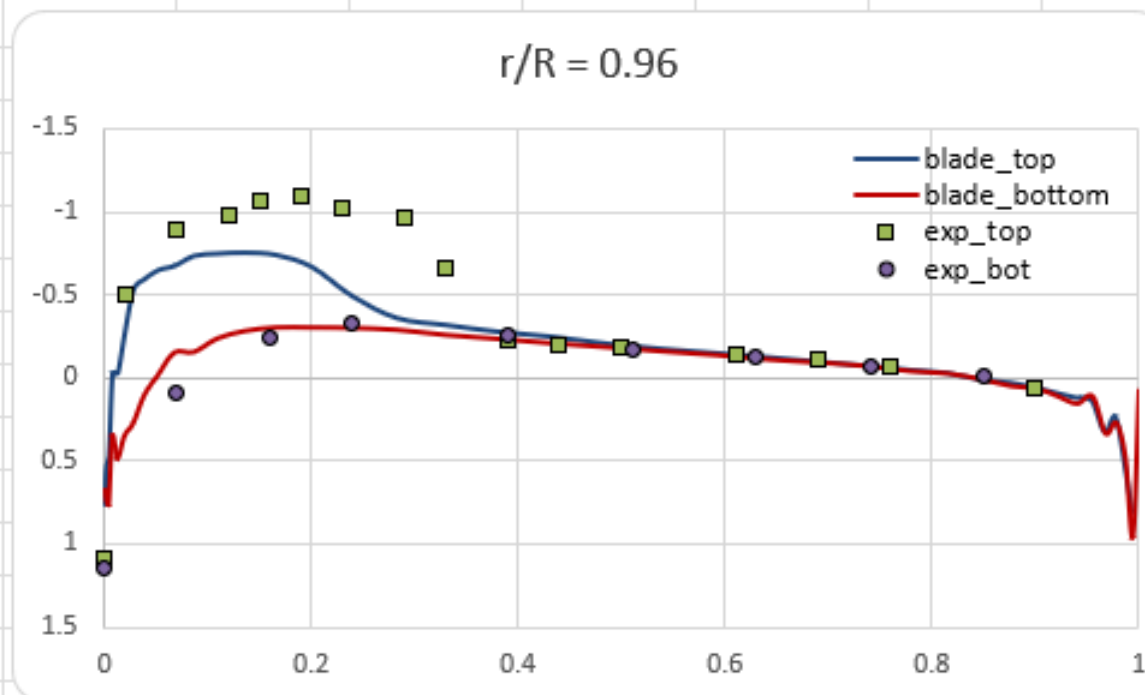
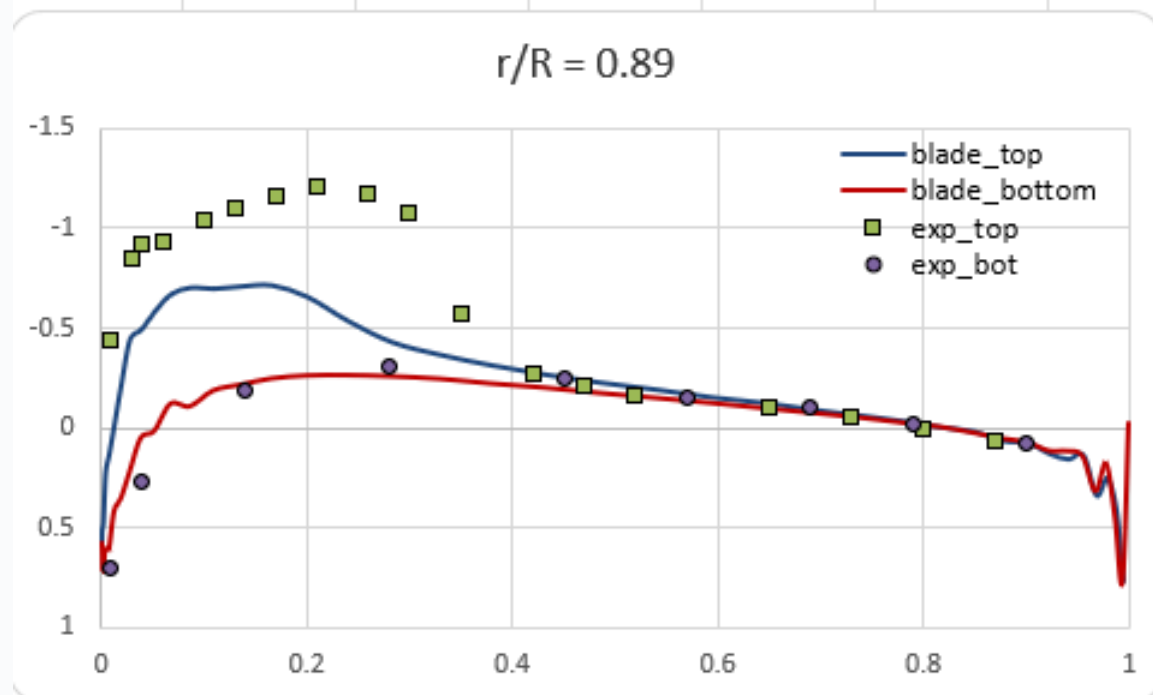
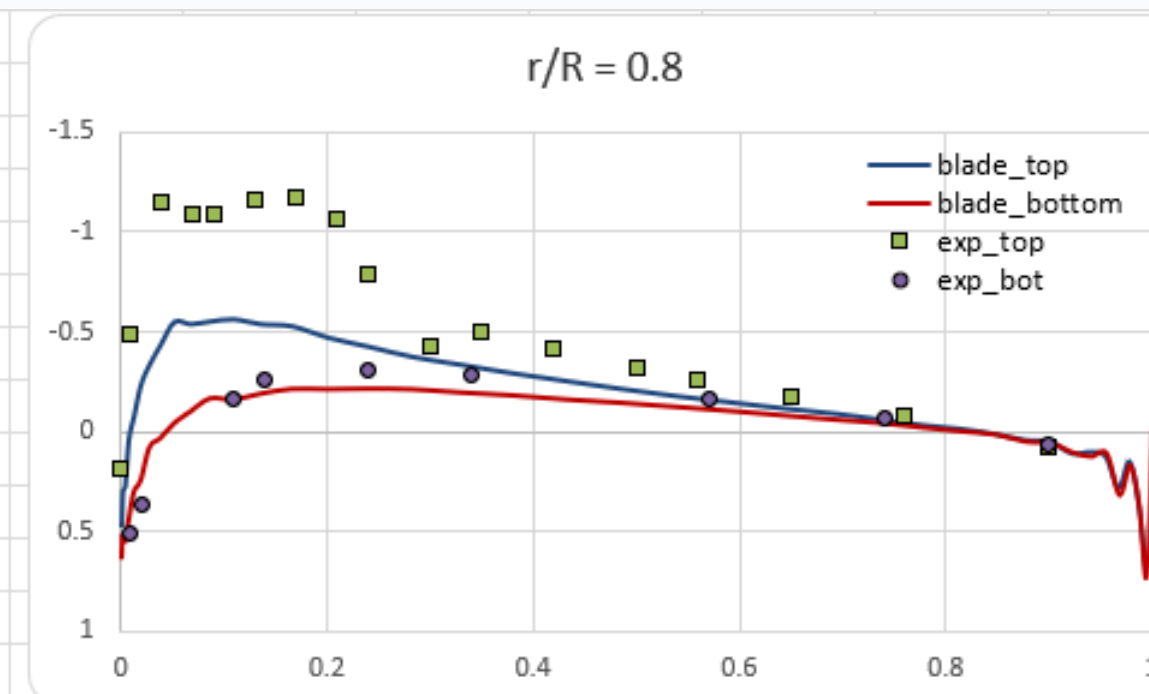
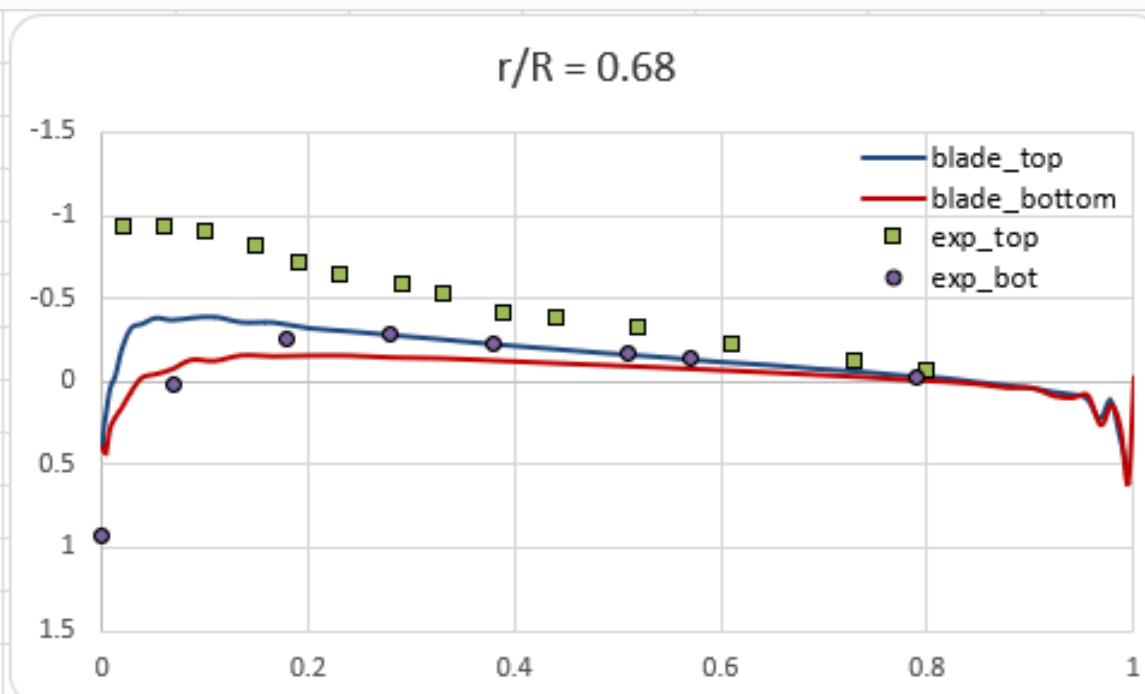
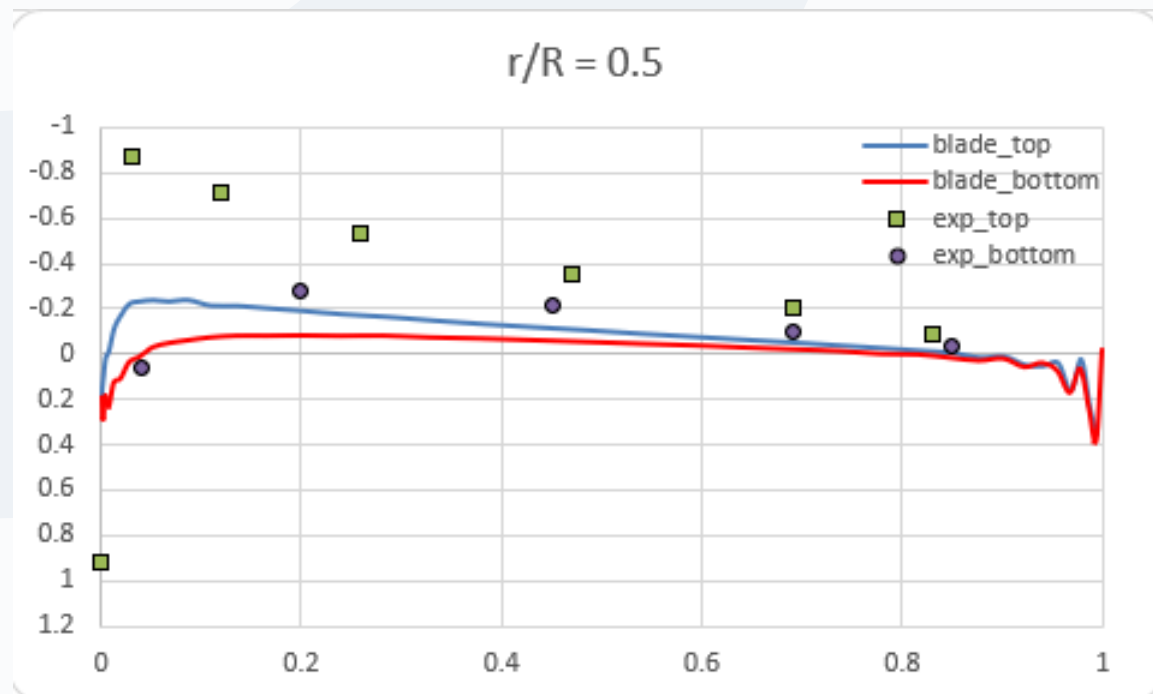
Upper surface ($-C_{pu}$)									
x/c	r/R = 0.5	x/c	r/R = 0.68	x/c	r/R = 0.80	x/c	r/R = 0.89	x/c	r/R = 0.96
0.00	-0.9138E+00	0.02	0.9416E+00	0.00	-0.1853E+00	0.01	0.4445E+00	0.00	-0.1092E+01
0.03	0.8691E+00	0.06	0.9399E+00	0.01	0.4832E+00	0.03	0.8531E+00	0.02	0.4984E+00
0.12	0.7173E+00	0.10	0.9083E+00	0.04	0.1152E+01	0.04	0.9171E+00	0.07	0.8911E+00
0.26	0.5317E+00	0.15	0.8169E+00	0.07	0.1091E+01	0.06	0.9325E+00	0.12	0.9721E+00
0.47	0.3517E+00	0.19	0.7256E+00	0.09	0.1095E+01	0.10	0.1046E+01	0.15	0.1062E+01
0.69	0.2055E+00	0.23	0.6448E+00	0.13	0.1166E+01	0.13	0.1108E+01	0.19	0.1098E+01
0.83	0.8736E-01	0.29	0.5834E+00	0.17	0.1170E+01	0.17	0.1166E+01	0.23	0.1028E+01
		0.33	0.5307E+00	0.21	0.1061E+01	0.21	0.1208E+01	0.29	0.9676E+00
		0.39	0.4183E+00	0.24	0.7839E+00	0.26	0.1172E+01	0.33	0.6605E+00
		0.44	0.3937E+00	0.30	0.4320E+00	0.30	0.1076E+01	0.39	0.2239E+00
		0.52	0.3235E+00	0.35	0.4990E+00	0.35	0.5771E+00	0.44	0.1947E+00
		0.61	0.2251E+00	0.42	0.4162E+00	0.42	0.2751E+00	0.50	0.1902E+00
		0.73	0.1338E+00	0.50	0.3230E+00	0.47	0.2111E+00	0.61	0.1474E+00
		0.80	0.6530E-01	0.56	0.2586E+00	0.52	0.1673E+00	0.69	0.1103E+00
				0.65	0.1746E+00	0.65	0.9980E-01	0.76	0.6527E-01
				0.76	0.7476E-01	0.73	0.5006E-01	0.90	-0.6074E-01
				0.90	-0.8021E-01	0.80	-0.6794E-02		
						0.87	-0.7194E-01		
Lower surface ($-C_{pl}$)									
x/c	r/R = 0.5	x/c	r/R = 0.68	x/c	r/R = 0.80	x/c	r/R = 0.89	x/c	r/R = 0.96
0.04	-0.5606E-01	0.00	-0.9269E+00	0.01	-0.5175E+00	0.01	-0.7021E+00	0.00	-0.1141E+01
0.20	0.2758E+00	0.07	-0.2059E-01	0.02	-0.3691E+00	0.04	-0.2745E+00	0.07	-0.8774E-01
0.45	0.2195E+00	0.18	0.2523E+00	0.11	0.1601E+00	0.14	0.1886E+00	0.16	0.2487E+00
0.69	0.9861E-01	0.28	0.2804E+00	0.14	0.2534E+00	0.28	0.3035E+00	0.24	0.3263E+00
0.85	0.3674E-01	0.38	0.2335E+00	0.24	0.3020E+00	0.45	0.2514E+00	0.39	0.2588E+00
		0.51	0.1679E+00	0.34	0.2862E+00	0.57	0.1543E+00	0.51	0.1755E+00
		0.57	0.1364E+00	0.57	0.1601E+00	0.69	0.9980E-01	0.63	0.1227E+00
		0.79	0.2036E-01	0.74	0.6557E-01	0.79	0.2163E-01	0.74	0.6640E-01
				0.90	-0.6445E-01	0.90	-0.7312E-01	0.85	0.1127E-01
C_L	0.2298	0.2842		0.2736		0.2989		0.3175	

해석조건

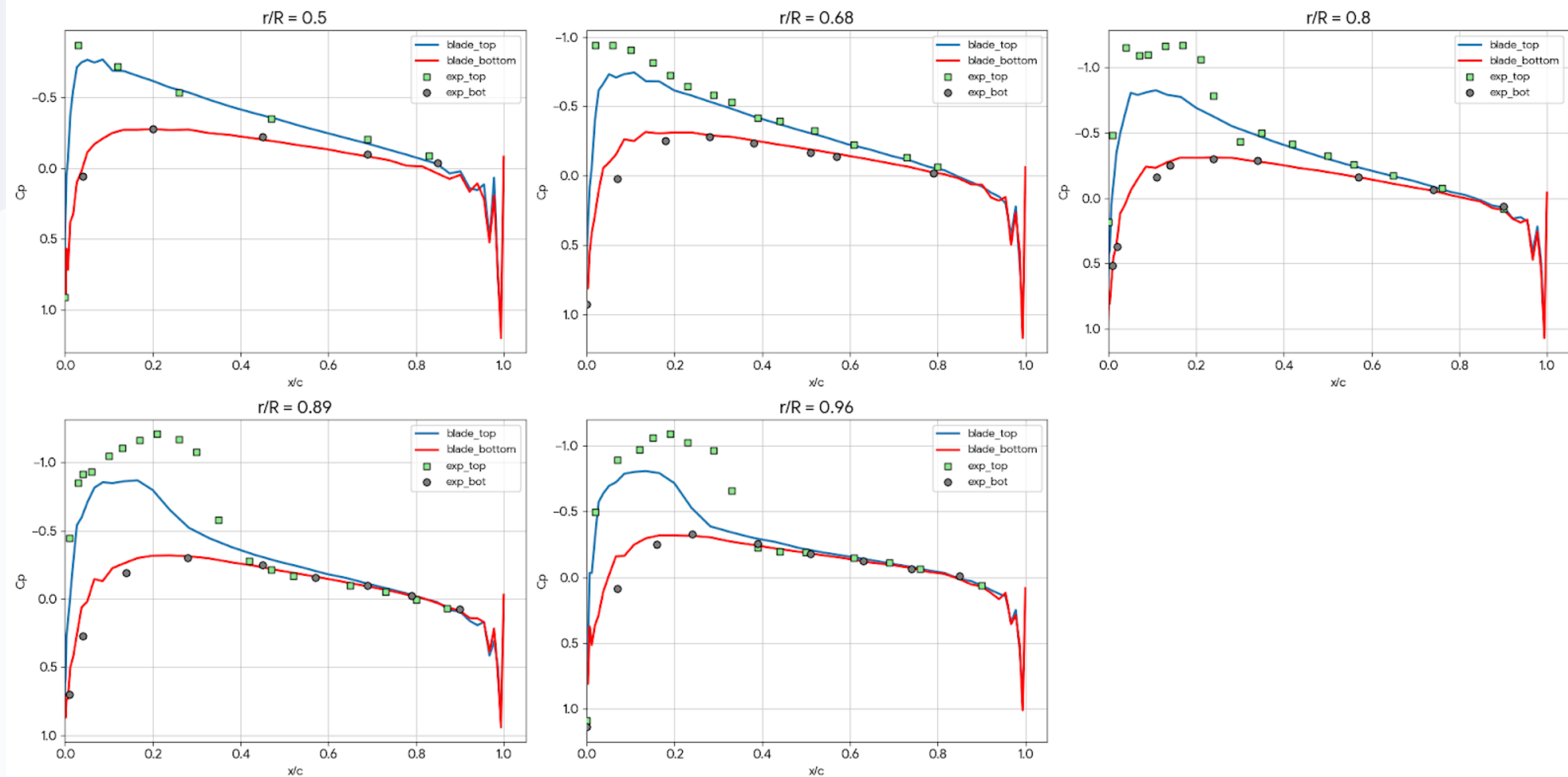
```
% ----- DYNAMIC MESH DEFINITION -----%  
%  
GRID_Movement= ROTATING FRAME  
MACH_MOTION= 0.877  
MOTION_ORIGIN= 0.0 0.0 0.0  
ROTATION_RATE = 261.79938779914943 0.0 0.0  
% 1250 RPM -> 130.89969389957471  
% 2500 RPM -> 261.79938779914943
```

피치각 8도는 격자에서 주어져 있음

해석값 실험값 비교 (파라뷰Cp)



해석값 실험값 비교 (압력계수 직접계산)



결과분석

초기 파라뷰를 통해 추출한 압력 계수(C_p) 데이터는 실험값과 큰 오차를 보였다. 원인 분석 결과, CFD 솔버는 전체 유동장에 대해 단일 기준 속도(Tip Speed 등)를 적용하여 C_p 를 산출하는 반면, 로터 블레이드는 반경 위치(r)에 따라 회전 속도($V=r*\Omega$)가 변하는 특성을 가진다.

따라서 본 연구에서는 각 섹션(Sec 1~5)의 반경 위치에 따른 국소 동압을 적용하여 C_p 를 재계산하였다. 그 결과, 섹션별 블레이드 윗면과 아랫면의 압력 분포가 실험 데이터(Caradonna & Tung)와 비교적 잘 일치함을 확인하였다.

참고문헌 그래프

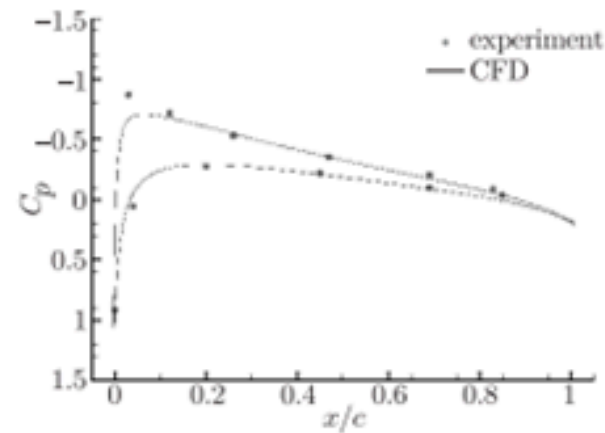


Figure 15. Pressure coefficient C_p distribution at radial location $r/R = 0.5$

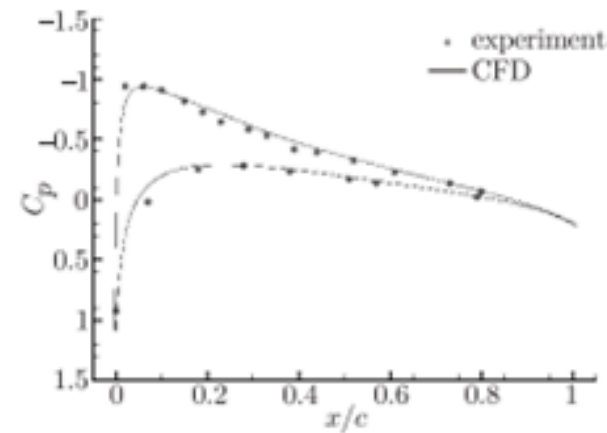


Figure 16. Pressure coefficient C_p distribution at radial location $r/R = 0.68$

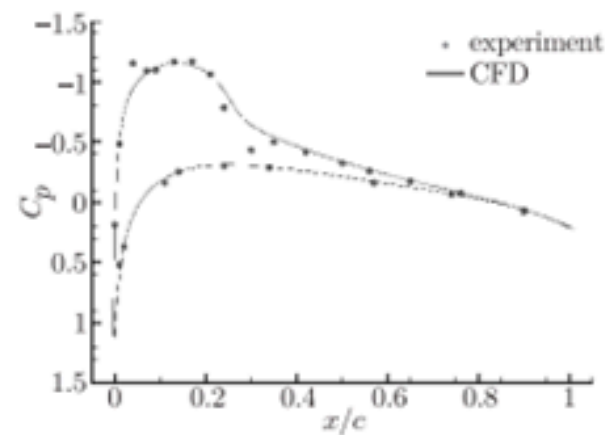


Figure 17. Pressure coefficient C_p distribution at radial location $r/R = 0.80$

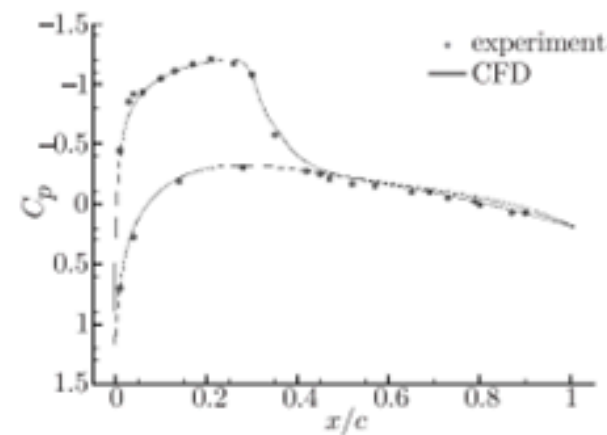
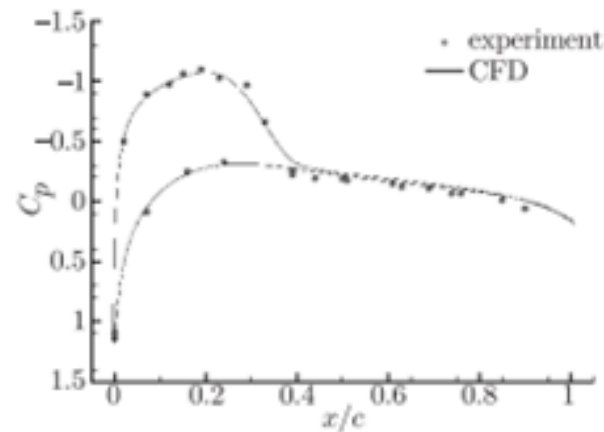
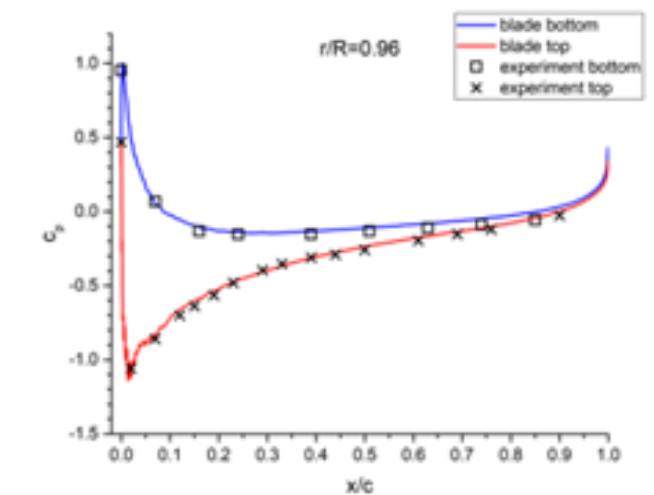
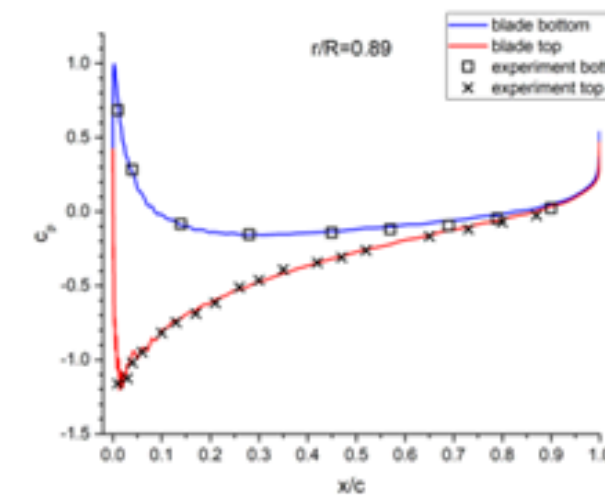
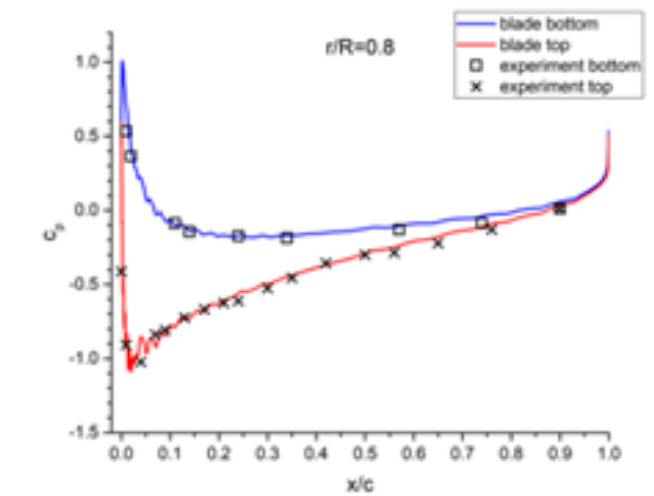
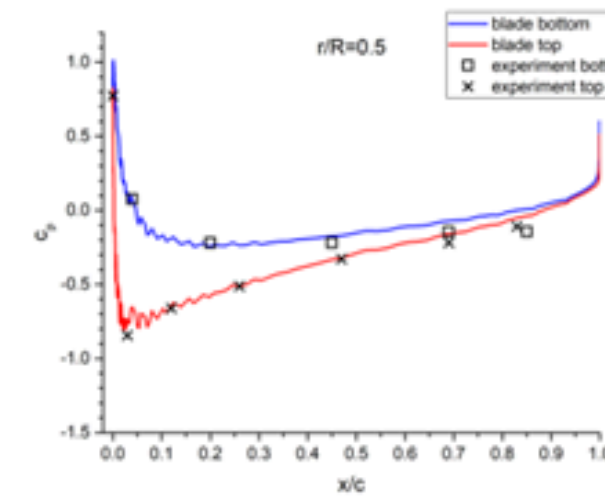


Figure 18. Pressure coefficient C_p distribution at radial location $r/R = 0.89$



RANS case: aerodynamics



NUMERICAL SIMULATION OF MODEL HELICOPTER ROTOR IN HOVER
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감사합니다

Thank you